Data Preprocessing

Steps Taken:

1. **Data Cleaning**
    - Converted TotalCharges to numeric
    - Dropped missing or erroneous values
    - Encoded target Churn to 0/1
2. **Feature Engineering**
    - tenure_group → Categorized tenure as Low / Medium / High
    - avg_monthly_spend → TotalCharges / (tenure + 1)
    - service_count → Count of services subscribed (e.g., Streaming TV, TechSupport)
    - engagement score → tenure * service_count
3. **Encoding Categorical Variables**
    - One-hot encoding for non-numeric columns (Contract, tenure_group, etc.)
4. **Handling Imbalanced Classes**
    - Applied **SMOTE** to oversample churned customers
5. **Scaling**
    - StandardScaler applied to numeric features for SVM, KNN, and Neural Network models

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